Work, Kinetic Energy & Gravitational Potential Energy

Answer key

Conventions used throughout

• **Work (from the lesson):** $W = (F \cos \theta)\Delta d$, where θ is the angle between the force and the displacement. Work is a scalar: it has a sign but no direction.

• **Energy symbols:** $E_k = \frac{1}{2}mv^2$ (kinetic), $\Delta E_g = mg\Delta y$ (gravitational potential). Work–energy theorem: $W_{net} = \Delta E_k$.

• **Signs:** work is positive when the force helps the motion (θ less than 90°) and negative when it opposes the motion (θ more than 90°).

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• **Final answers:** work and energy are scalars (a size and a sign, no direction). Forces and velocities are given as a size with the direction in square brackets.

• $g = 9.81\ m/s^2$. Speeds in km/h are changed to m/s (÷ 3.6). Every step keeps full precision; only final answers are rounded.

WORK AND ENERGY 1

Force with zero work

1. Describe a situation where a force is exerted, but zero work is being done. Based on the above relationship, what angle must the force be applied to result in zero work?

Work is $W = (F \cos \theta)\Delta d$. It is zero when the force has no component along the displacement, or when there is no displacement at all.

Formula: $W = (F\cos\theta)\,\Delta d$

Zero work with F and Δd not zero: $\cos\theta = 0 \quad \Rightarrow \quad \theta = 90°$

Example: you carry a backpack across a level floor at constant velocity. Your force on the backpack is straight up and the displacement is horizontal, so θ = 90° and you do no work on it. (Pushing on a wall that does not move is another case: Δd = 0, so W = 0.)

FINAL ANSWER

A force perpendicular to the displacement does zero work, for example the upward force on a backpack carried horizontally.

The force must be applied at θ = 90° to the displacement.

WORK AND ENERGY 2

The joule

2. Recall that work is measure in Joules. Based on the equation, express Joules in terms of meters and Newtons

Put the units into W = (F cos θ)Δd. The cosine has no units.

Formula: $W = (F\cos\theta)\,\Delta d$

Units: $[\mathrm{J}] = [\mathrm{N}]\,[\mathrm{m}]$

FINAL ANSWER

1 J = 1 N·m (one newton times one metre)

EXAMPLE

Positive and negative work

Rony pulls a 15.0 kg wagon with an applied force of 25.0 N [30.0^{o} above the horizontal] over a distance of 10.0 m. The wagon experiences a force of kinetic friction of 15.0 N.

(a) How much work does Rony do on the wagon?

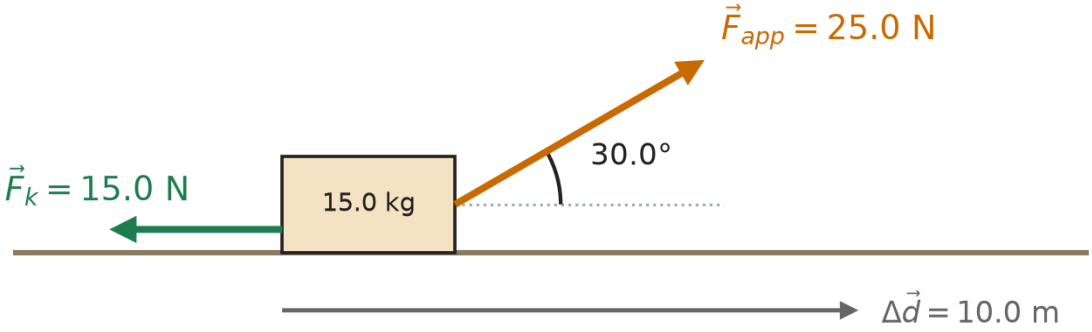
(b) How much work does friction do on the wagon?

(c) What is the meaning of the minus sign in part (b)?

GIVEN

$m = 15.0$ kg	$F_{app} = 25.0$ N	$\theta = 30.0°$
$\Delta d = 10.0$ m	$F_k = 15.0$ N	$W = ?$

The wagon: applied force, friction and displacement



(a) How much work does Rony do on the wagon?

Only the part of Rony's force that is parallel to the displacement does work. That part is $F_{app} \cos \theta$.

Formula: $W = (F\cos\theta)\,\Delta d$

Put in the numbers:

$$W_{app} = (25.0\cos 30.0°)(10.0) = (21.651)(10.0) = 216.51 \text{ J}$$

(b) How much work does friction do on the wagon?

Kinetic friction points backward, opposite to the displacement, so the angle between them is 180°.

Formula: $W = (F\cos\theta)\,\Delta d$

Put in the numbers:

$$W_f = (15.0\cos 180°)(10.0) = (-15.0)(10.0) = -150.0\ \text{J}$$

(c) What is the meaning of the minus sign in part (b)?

Friction acts opposite to the displacement, so it takes energy away from the wagon instead of giving it energy. The 150 J that friction removes becomes thermal energy in the wheels and the ground.

FINAL ANSWER

(a) $W_{app} = 217$ J

(b) $W_f = -1.50 \times 10^2$ J

(c) Friction opposes the motion, so it removes energy from the wagon

Worksheet answer: (a) 216 J
Why the answers differ: Same working, different rounding. (25.0 cos 30.0°)(10.0) = 216.51 J, which rounds to 217 J at 3 significant figures. The worksheet cut the number off at 216 instead of rounding it.

EXAMPLE

Lifting

Consider Jennifer lifting a 12 kg box to a height of 1.3 m at a constant speed.

(a) Draw a free-body diagram for the situation.

(b) What is the magnitude of the force required to lift the box at constant speed and the magnitude of the work done to lift it?

(c) Determine the work done by gravity during the lift.

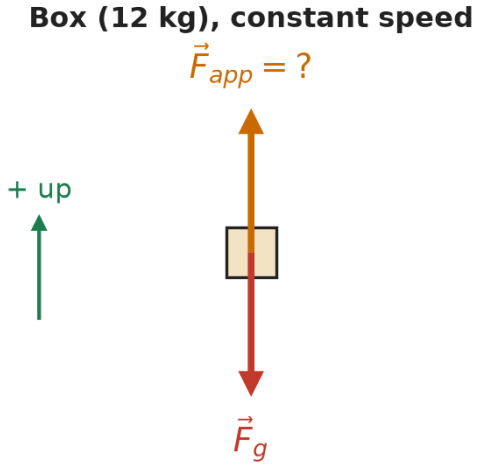
GIVEN

$m = 12$ kg $\quad \Delta y = 1.3$ m $\quad F_{app} = ?$ $\quad W = ?$

(a) Draw a free-body diagram for the situation.

Two forces act on the box: Jennifer's applied force up and gravity down. The speed is constant, so the acceleration is zero and the two arrows are the same length.

FBD of the box (+y up)



(b) What is the magnitude of the force required to lift the box at constant speed and the magnitude of the work done to lift it?

Step 1 – Applied force (vertical balance)

Constant speed means the net force is zero. Forces are written as sizes, with + for up:

Balance: $F_{net,y} = 0, \quad \text{so} \quad 0 = F_{app} - F_g$

Add F_g to both sides: $F_{app} = F_g = mg$

Put in the numbers: $F_{app} = (12)(9.81) = 117.72 \text{ N}$

Step 2 – Work done by Jennifer

Her force is up and the displacement is up, so θ = 0° and cos 0° = 1.

Formula: $W = (F\cos\theta)\,\Delta d$

Put in the numbers: $W_{app} = (117.72)(\cos 0°)(1.3) = 153.04 \text{ J}$

(c) Determine the work done by gravity during the lift.

Gravity points down and the displacement is up, so θ = 180° and cos 180° = −1.

Formula: $W = (F\cos\theta)\,\Delta d$

Put in the numbers: $W_g = (117.72)(\cos 180°)(1.3) = -153.04\ \text{J}$

The two works add to zero, which matches the constant speed: the box's kinetic energy does not change.

FINAL ANSWER

(a) F_{app} up and F_g down, equal in length

(b) $F_{app} = 1.2 \times 10^2$ N; $W_{app} = 1.5 \times 10^2$ J

(c) $W_g = -1.5 \times 10^2$ J

Worksheet answer: 118 N; 153 J; −153 J
Why the answers differ: Same values, different rounding. (12)(9.81) = 117.72 N and (117.72)(1.3) = 153.04 J. The data (12 kg and 1.3 m) have 2 significant figures, so the key gives 1.2×10^2 N and 1.5×10^2 J; the worksheet keeps 3.

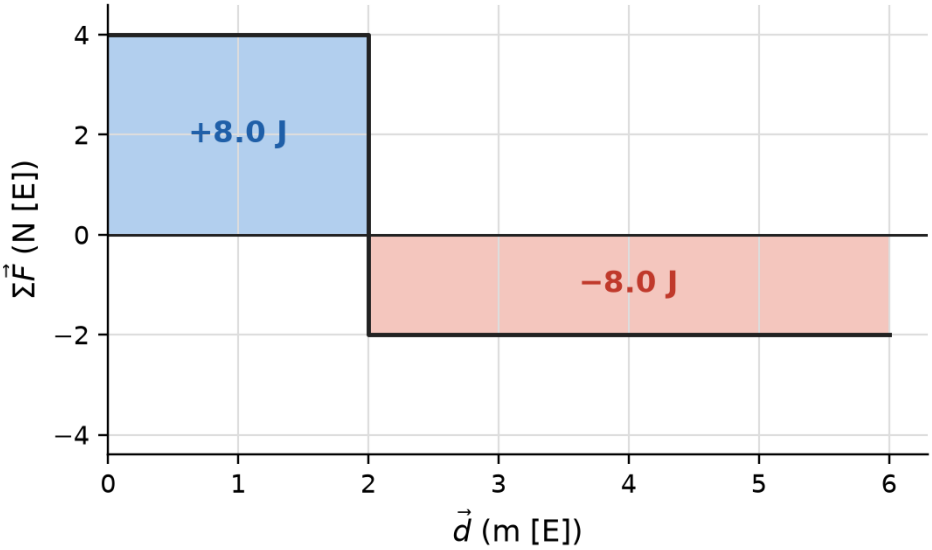
FORCE-DISPLACEMENT GRAPHS

Area under the graph

The graph on the right represents the net horizontal forces acting on an object as a function of displacement along a horizontal surface.

1. Determine the area of the first section for a displacement from 0 to 2.0 m [E]. What does this area represent? (Hint: Consider units)
2. Determine the total area up to 6.0 m [E].
3. Describe a physical situation that could result in this graph.

The worksheet's graph, with the two areas marked



Read from the graph: ΣF = 4.0 N [E] from 0 to 2.0 m, then ΣF = −2.0 N (2.0 N [W]) from 2.0 m to 6.0 m.

1. Determine the area of the first section for a displacement from 0 to 2.0 m [E]. What does this area represent? (Hint: Consider units)

The first section is a rectangle: height 4.0 N, width 2.0 m.

Area of a rectangle: $A_1 = (\text{height})(\text{width})$

Put in the numbers: $A_1 = (4.0\ \text{N})(2.0\ \text{m}) = 8.0\ \text{N·m} = 8.0\ \text{J}$

The units are N·m = J, the units of work. The area under a force–displacement graph is the work done on the object, here the net work.

2. Determine the total area up to 6.0 m [E].

The second section is below the axis, so its area counts as negative: height −2.0 N, width 6.0 m − 2.0 m = 4.0 m.

$$\text{Second area:}\quad A_2 = (-2.0\ \text{N})(4.0\ \text{m}) = -8.0\ \text{J}$$

$$\text{Total area:}\quad A = A_1 + A_2 = 8.0 + (-8.0) = 0\ \text{J}$$

The net work over the whole 6.0 m is zero, so the object ends with the same kinetic energy it started with.

3. Describe a physical situation that could result in this graph.

A box at rest on a floor is pushed east so that the net force on it is 4.0 N [E] for the first 2.0 m, and it speeds up. The push then stops, and friction (2.0 N [W]) is the only horizontal force. The box slows down over the next 4.0 m and comes to rest at 6.0 m, because the 8.0 J it gained has all been removed.

FINAL ANSWER

1. Area = 8.0 J. It is the (net) work done on the object.

2. Total area = 0 J

3. For example: a box pushed from rest for 2.0 m, then slowed to a stop by friction over the next 4.0 m

EXAMPLE 1

Calculating kinetic energy

A car has a mass of 1.50 x 10^3 kg and is travelling at a speed of 85.0 km/h. Calculate the car's kinetic energy.

GIVEN

$m = 1.50 \times 10^3$ kg $\qquad v = 85.0$ km/h $\qquad E_k = ?$

Step 1 – Change the speed to m/s

Divide by 3.6: $v = \frac{85.0}{3.6} = 23.611 \text{ m/s}$

Step 2 – Kinetic energy

Formula: $E_k = \frac{1}{2}mv^2$

Put in the numbers:

$$E_k = \frac{1}{2}(1.50 \times 10^3)(23.611)^2 = 4.181 \times 10^5 \text{ J}$$

FINAL ANSWER

$E_k = 4.18 \times 10^5$ J

EXAMPLE 2

Applying the work-energy theorem

A blue whale with a mass of 1.5 x 10^5 kg is swimming with a speed of 6.1 m/s. A nearby boat startles the whale, and the whale increases its speed to 12.8 m/s. Calculate the work done on the whale by the water.

GIVEN

$m = 1.5 \times 10^5$ kg

$v_1 = 6.1$ m/s

$v_2 = 12.8$ m/s

$W = ?$

The work done on the whale only changes its speed, so it equals the change in the whale's kinetic energy (work–energy theorem).

Formula: $W = \Delta E_k = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$

Factor out $\frac{1}{2}m$: $W = \frac{1}{2}m\left(v_2^2 - v_1^2\right)$

Put in the numbers: $W = \frac{1}{2}(1.5 \times 10^5)(12.8^2 - 6.1^2)$

$W = \frac{1}{2}(1.5 \times 10^5)(163.84 - 37.21) = 9.497 \times 10^6 \text{ J}$

Why the water: the whale pushes backward on the water with its tail, and by Newton's third law the water pushes forward on the whale. That forward force from the water is the outside force that acts on the whale through a distance, so the water does the work.

FINAL ANSWER

W = 9.5 × 10^6 J

EXAMPLE 3

Rony & the wagon from page 1

Recall that on page 1, Rony did 216.5 J of work on the wagon while friction did -150 J of work.

a) What is the net work done on the wagon?

b) If the wagon had an initial velocity of 1.2 m/s and a mass of 5.0 kg, how fast was it going after Rony pulled it for 10.0 m?

GIVEN

$W_{app} = 216.5 \text{ J}$	$W_f = -150 \text{ J}$	$v_1 = 1.2 \text{ m/s}$
$m = 5.0 \text{ kg}$	$W_{net} = ?$	$v_2 = ?$

a) What is the net work done on the wagon?

Work is a scalar, so the net work is the sum of the works, each with its sign. (Gravity and the normal force are perpendicular to the motion and do no work.)

Formula: $W_{net} = W_{app} + W_f$

Put in the numbers: $W_{net} = 216.5 + (-150) = 66.5 \text{ J}$

b) If the wagon had an initial velocity of 1.2 m/s and a mass of 5.0 kg, how fast was it going after Rony pulled it for 10.0 m?

The net work equals the change in kinetic energy. Isolate v_2 in letters first. (This part uses the mass it gives, 5.0 kg.)

Formula: $W_{net} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$

Add $\frac{1}{2}mv_1^2$, then multiply by $\frac{2}{m}$: $v_2^2 = v_1^2 + \frac{2W_{net}}{m}$

Square root both sides: $v_2 = \sqrt{v_1^2 + \frac{2W_{net}}{m}}$

Put in the numbers:

$$v_2 = \sqrt{(1.2)^2 + \frac{2(66.5)}{5.0}} = \sqrt{28.04} = 5.295 \text{ m/s}$$

FINAL ANSWER

a) W_{net} = +67 J

b) v_2 = 5.3 m/s

Worksheet answer: a) +66 J
Why the answers differ: Different rounding. 216.5 − 150 = 66.5 J, and the unrounded value from page 1 is 216.51 − 150 = 66.51 J, which rounds up to 67 J. The worksheet cut it off at 66 J. Part b) is not affected.

EXAMPLE 3

Applications involving gravitational potential energy

Yan stands near the edge of a cliff and accidentally drops a rock of mass 1.2 kg to the ground at the base of the cliff 28 m below. Calculate the potential energy of the rock relative to the ground just before he drops it.

GIVEN

$m = 1.2$ kg $\qquad \Delta y = 28$ m $\qquad E_g = ?$

"Relative to the ground" means the ground is the reference level (y = 0), so the rock is 28 m above it.

Formula: $\Delta E_g = mg\,\Delta y$

Put in the numbers: $E_g = (1.2)(9.81)(28) = 329.62$ J

FINAL ANSWER

$\mathbf{E_g = 3.3 \times 10^2\ J}$

EXAMPLE 4

Applying gravitational potential energy to determine mass

Jack raises a loaded barbell 2.2 m. The lift increases the gravitational potential energy of the barbell by 490 J. Determine the mass of the loaded barbell.

GIVEN

$\Delta y = 2.2$ m $\qquad \Delta E_g = 490$ J $\qquad m = ?$

Isolate m in letters first, then put in the numbers.

Formula: $\Delta E_g = mg\,\Delta y$

Divide both sides by $g\,\Delta y$: $m = \dfrac{\Delta E_g}{g\,\Delta y}$

Put in the numbers: $m = \dfrac{490}{(9.81)(2.2)} = \dfrac{490}{21.582} = 22.70 \text{ kg}$

FINAL ANSWER

m = 23 kg